

CS 230 : Discrete Computational Structures

Fall Semester, 2023

HOMEWORK ASSIGNMENT #1

Due Date: Tuesday, September 5

Suggested Reading: Rosen Sections 1.1 - 1.3; LLM Sections 1.1, 3.1 - 3.4

These are the problems that you need to hand in for grading. Always explain your answers and show your reasoning.

1. **[6 Pts]** Translate the following English sentences into logic. First, define your basic propositions and use logical operations to connect them.
 - (a) If n is prime, then $n = 2$ or n is odd.
 - (b) The differentiability of f is sufficient for f to be continuous and smooth.
 - (c) Being a good programmer and having good communication skills are necessary to get a good job.
2. **[6 Pts]** Determine whether $((p \rightarrow q) \wedge (q \rightarrow r)) \leftrightarrow (p \rightarrow r)$ is a tautology using truth tables.
3. **[6 Pts]** Prove that $(p \rightarrow (q \rightarrow r))$ and $(q \rightarrow (p \rightarrow r))$ are logically equivalent by deduction using a series of logical equivalences studied in class (truth tables will not be allowed).
4. **[10 Pts]** Use logical reasoning to solve the following puzzle:
A detective has interviewed four witnesses to a crime. From the stories of the witnesses the detective has concluded that if the butler is telling the truth then so is the cook. The cook and the gardener are either both telling the truth or they are both lying. Either the gardener or the handyman is telling the truth, but not both. If the handyman is lying then so is the cook. Determine which witnesses are lying and which are telling the truth. Define four basic propositions and use these to describe the four statements above. Then use deductive reasoning, not truth tables, to derive the truth of each of those basic propositions.
5. **[6 Pts]** A proposition is said to be in CNF (conjunctive normal form) if it is a conjunction (and) of one or more clauses, where each clause is a disjunction (or) of basic propositions or their negations. For example, $(p \vee \neg q) \wedge (\neg p \vee \neg q \vee r) \wedge (q \vee r)$ is in CNF. Now argue that any compound proposition is logically equivalent to some proposition in CNF and show your construction. *Hint: Look at the argument on DNF in the notes and come up with some sort of algorithm. Start by constructing a truth table.*
6. **[6 Pts]** Prove that $\{\vee, \oplus, \text{TRUE}\}$ is functionally complete, i.e., any propositional formula is equivalent to one whose only connectives are $\vee$, $\oplus$, and the constant TRUE. Prove using a series of logical equivalences. You may assume any logical equivalences

we studied in class and the fact that any formula is equivalent to some formula in DNF.

Note: If you make the statement that a set of operators is functionally complete, and use this in your proof, then you need to justify your statement.

For more practice, you are encouraged to work on the problems given at the end of Rosen, Sections 1.1 - 1.3.